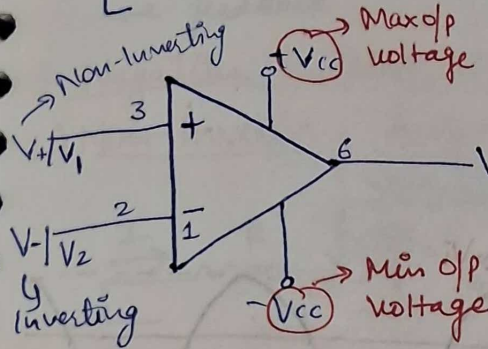


OP-Amp

- high voltage gain, direct coupled amplifier.
- voltage controlled voltage source ($V_c V_s$)
- IC-741

$$\left[\text{Voltage Gain}(A_v) = \frac{\text{O/P Voltage}}{\text{I/P Voltage}} \right]$$



- 8 Pins
- 1 & 5 → Offset, Nulls
- 8 → not connected
- 7 & 4 → biasing
- $V_d = V_1 - V_2$
- $V_+ - V_-$

- +ve feedback → $V_d \uparrow \rightarrow$ saturation region
- -ve feedback → $V_d \downarrow \rightarrow$ linear region

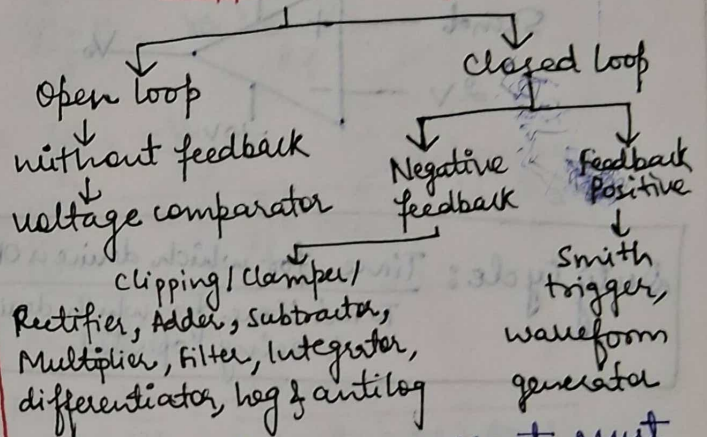
Virtual Ground

$$V_d \downarrow \rightarrow (V_u) \approx 0$$

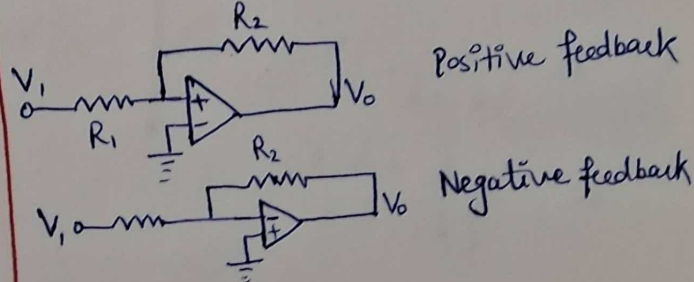
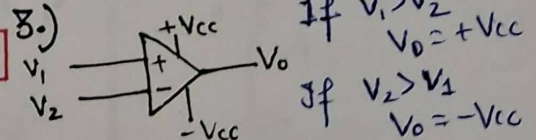
$$V_+ - V_- \approx 0 \rightarrow V_+ \approx V_-$$

→ only applicable in -ve feedback

Applications of op-amp



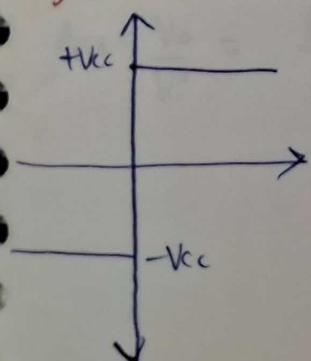
R_i should be $\uparrow$ bcz no current must flow into the input terminal of op-amp.
 $\rightarrow R_o \rightarrow \downarrow \rightarrow$ output voltage of op-amp must be independent of Load Resistor.



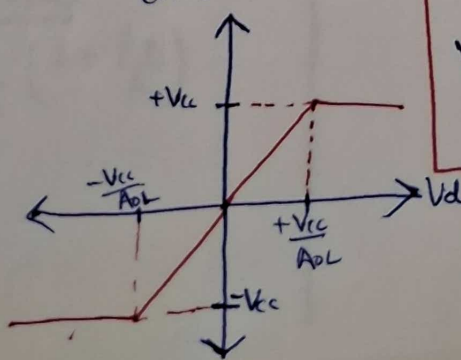
Properties	Ideal	Practical
a) A_{OL}	∞	10^6
b) R_i	∞	1M Ω -2M Ω
c) R_o	0	50 Ω -100 Ω
d) Open loop Bandwidth	∞	5Hz
e) CMRR	∞	10 6 to 120dB
f) Slew rate	∞	0.5 to 1V/ μ sec

Transfer ch

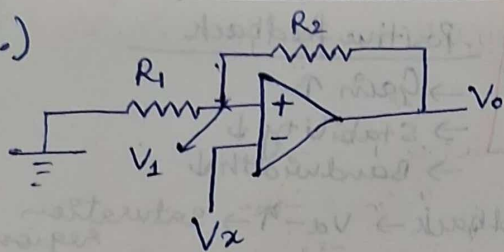
a) Ideal ($A_{OL} \rightarrow \infty$)



b) Practical ($A_{OL} \approx 10^4 - 10^6$)



5.)

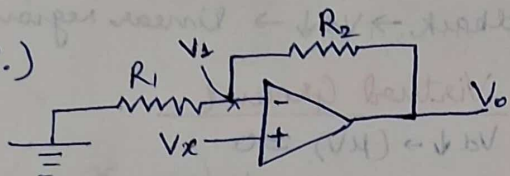


$$V_{\perp} = \left(\frac{R_1}{R_1 + R_2} \right) V_0$$

$$V_0 = A_{OL} (V_+ - V_-)$$

$$V_0 = A_{OL} V_d$$

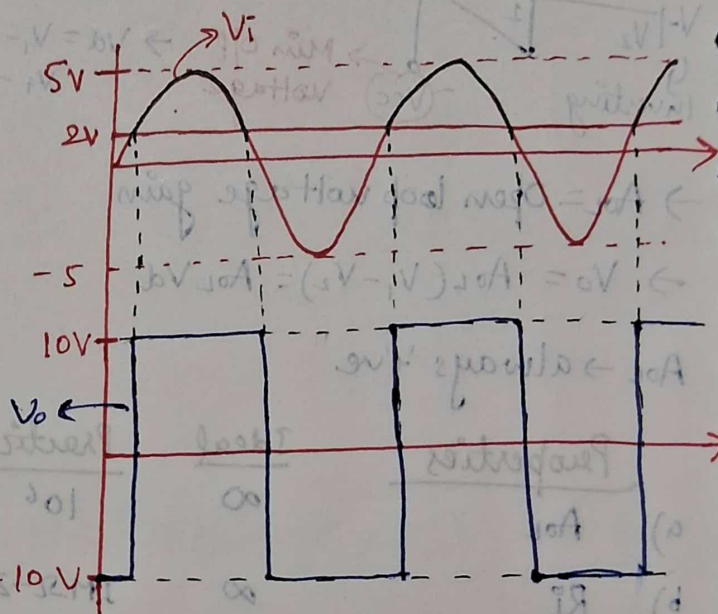
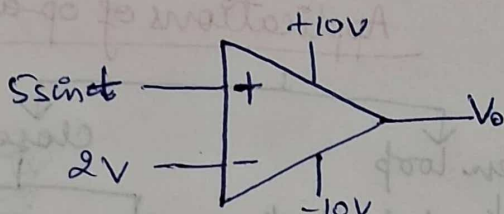
6.)



$$V_0 = A_{OL} (V_x - V_1)$$

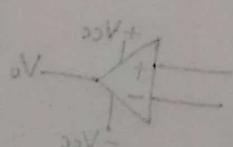
$$= A_{OL} V_d$$

Q Draw the output waveform

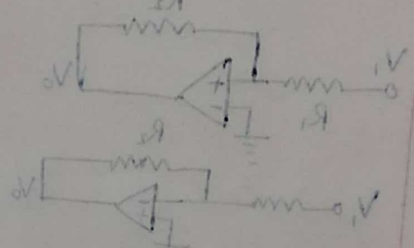


Duty Cycle: $\frac{\text{Time for which device is ON}}{\text{Total time for which device is being operated}}$

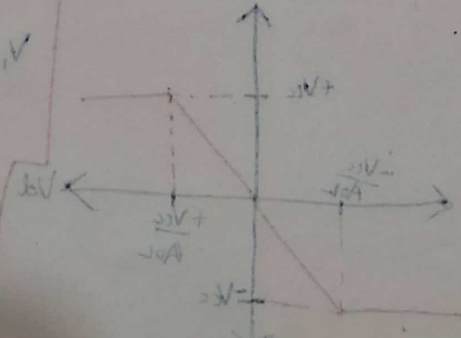
If $V_1 > V_2$
 $V_0 = +V_{CC}$
 If $V_1 < V_2$
 $V_0 = -V_{CC}$



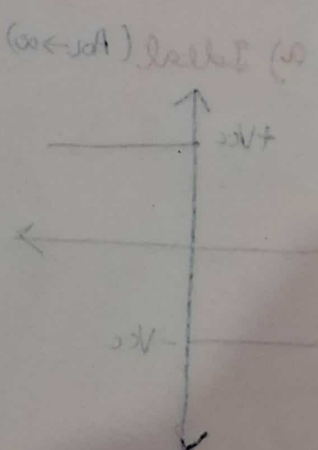
Positive feedback
 Negative feedback



limitation (d)
 $(A_{OL} = 10^4 - 10^5)$



Transfer for ch



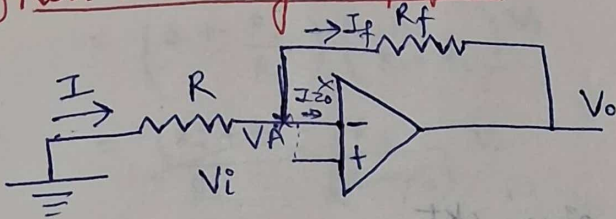
Assumptions for negative feedback

- 1.) Inverting & non-inverting nodes of op-amp are virtually short.
- 2.) Input current is zero
 ↓
 valid for both +ve & -ve feedback
 ↓
 valid for -ve feedback only

- Offset voltage, slew rate, CMRR
- zero crossing detector
- summing and difference amp
- unity gain buffer
- Integrator, Differentiator
- log amplifier

Negative feedback applications of op-amp

1.) Non-Inverting Amplifier



$$V_A = V_i \rightarrow \text{virtually short}$$

$$\rightarrow I_R = I_f$$

$$\rightarrow \frac{0 - V_A}{R} = \frac{V_A - V_o}{R_f}$$

$$\text{since } V_A = V_i$$

$$\Rightarrow -\frac{V_i}{R} = \frac{V_i - V_o}{R_f}$$

$$\Rightarrow -\frac{R_f}{R} = \frac{V_i - V_o}{V_i}$$

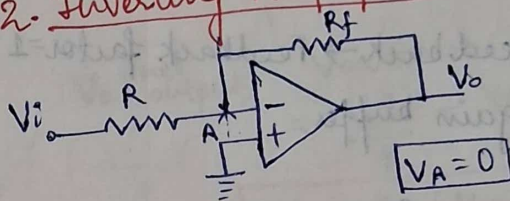
$$\Rightarrow 1 - \frac{V_o}{V_i} = -\frac{R_f}{R}$$

$$\Rightarrow 1 + \frac{R_f}{R} = \frac{V_o}{V_i} = A_{CL}$$

↓
Only applicable when $A_{OL} \rightarrow \infty$
Because Actually

$$A_{CL} = \frac{V_o}{V_i} = \frac{(1 + R_f/R)}{1 + \frac{1}{A_{OL}} \left(1 + \frac{R_f}{R}\right)}$$

2. Inverting Amplifier



$$I_R = I_f$$

$$\Rightarrow \frac{V_i - 0}{R} = \frac{V_A - V_o}{R_f}$$

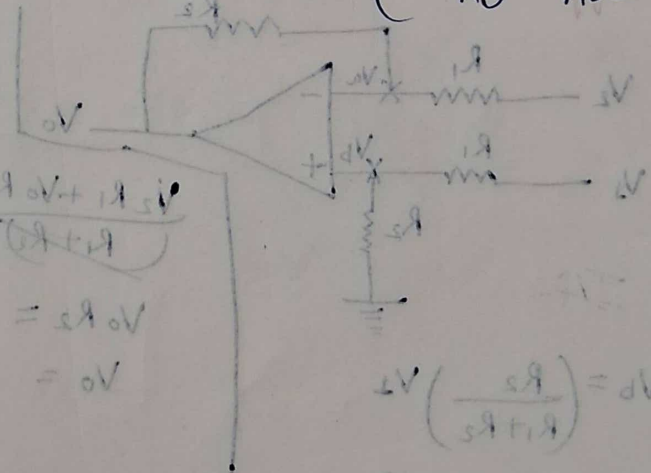
$$\Rightarrow \frac{V_i}{R} = \frac{-V_o}{R_f}$$

$$\Rightarrow \boxed{\frac{V_o}{V_i} = -\frac{R_f}{R} = A_{CL}}$$

only when $A_{OL} \rightarrow \infty$

if $A_{OL} \rightarrow \text{finite}$

$$A_{CL} = \frac{V_o}{V_i} = \frac{-(R_f/R)}{1 + \left(1 + \frac{R_f}{R}\right) \frac{1}{A_{OL}}}$$

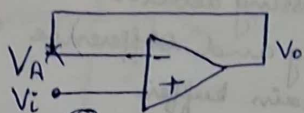


$$1V \left(\frac{5k}{5k+1k} \right) = 0V$$

$$5kV + 1kV = 0V$$

$$\frac{5kV + 1kV}{(1k+5k)}$$

3.) Voltage Follower/Buffer



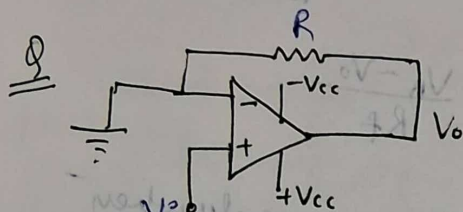
$V_A = V_i$ (Virtual Short) & $V_A = V_o$ (Physical short)

$\rightarrow V_o = V_i$ (from ① and ②)

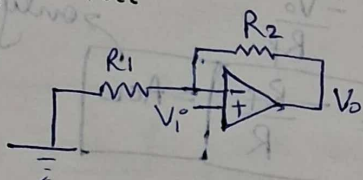
$\rightarrow \frac{V_o}{V_i} = 1$ (A_{CL})

$\rightarrow 100\%$ feedback $\rightarrow$ Feedback factor = 1

$\rightarrow$ Unity gain buffer



similar to



For the given ckt

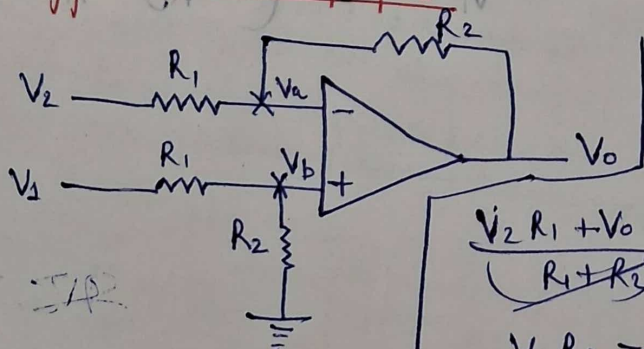
$R_1 = 0$

$V_o = \left(1 + \frac{R_2}{R_1}\right) V_i$

$= \left(1 + \frac{R}{0}\right) V_i$

$= \infty \approx +V_{CC}$

4.) Differential Amplifier



$V_a = V_o$

$V_2 R_1 + V_o R_2$

$\frac{V_2 R_1 + V_o R_2}{(R_1 + R_2)} = \frac{R_2 V_1}{(R_1 + R_2)}$

$V_o R_2 = R_2 V_1 - V_2 R_1$

$V_o = \frac{R_2 V_1 - V_2 R_1}{R_2}$

a) $V_b = \left(\frac{R_2}{R_1 + R_2}\right) V_1$

$V_a = \frac{V_2 R_1 + V_o R_2}{(R_2 + R_1)}$

$V_2 - V_a - \frac{R_1 - V_a - (V_a - V_o) R_2}{R_2} = V_o$

$V_2 - V_a - \frac{V_a - V_a + V_o - V_o}{R_2} = V_o$

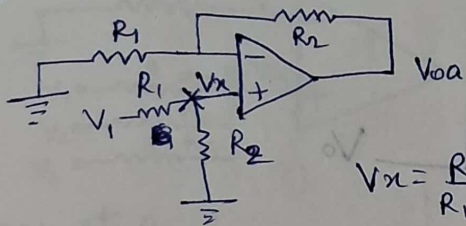
$V_2 - 3V_a = 0$

$10A = \frac{0V}{9V} = \frac{9R + 1}{R}$

$\frac{(9R + 1)}{(9R + 1)} \cdot \frac{1}{10A} + 1 = \frac{0V}{9V}$

b) Superposition Theorem

a) let V_1 be present & $V_2 = 0$



$$V_x = \frac{R_2}{R_1 + R_2} V_1$$

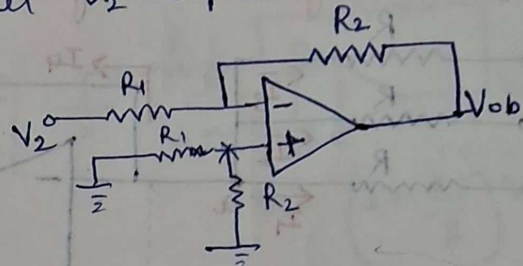
$$V_{oa} = \left(1 + \frac{R_2}{R_1}\right) V_x$$

$$= \left(1 + \frac{R_2}{R_1}\right) \left(\frac{R_2}{R_1 + R_2}\right) V_1$$

$$= \frac{(R_1 + R_2)}{R_1} \cdot \frac{R_2}{(R_1 + R_2)} V_1$$

$$V_{oa} = \frac{V_1 R_2}{R_1}$$

b) let V_2 be present & $V_1 = 0$



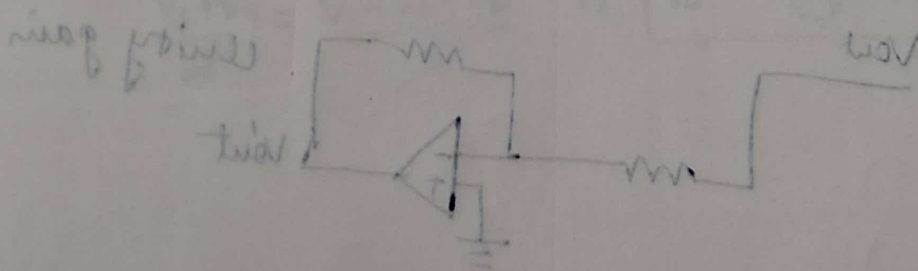
$$V_{ob} = -\frac{R_2}{R_1} V_2$$

$$V_o (\text{net output}) = V_{oa} + V_{ob}$$

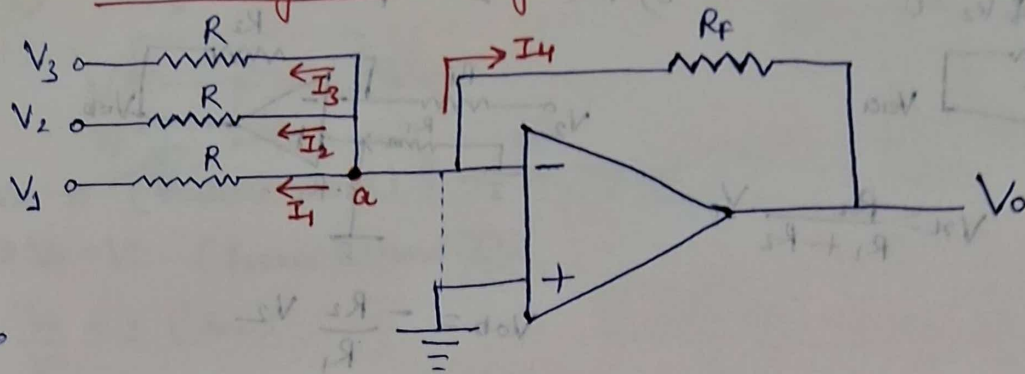
$$= \frac{V_1 R_2}{R_1} - \frac{V_2 R_2}{R_1}$$

$$V_o = \frac{R_2}{R_1} (V_1 - V_2)$$

to make positive or non-inverting



5.) Inverting Summing Amplifier



KCL at node A

$$I_1 + I_2 + I_3 + I_4 = 0$$

$$\Rightarrow \frac{V_a - V_1}{R} + \frac{V_a - V_2}{R} + \frac{V_a - V_3}{R} + \frac{V_a - V_o}{R_f} = 0$$

$V_a = 0$ (Virtual ground)

$$\Rightarrow -\frac{V_1}{R} - \frac{V_2}{R} - \frac{V_3}{R} - \frac{V_o}{R_f} = 0$$

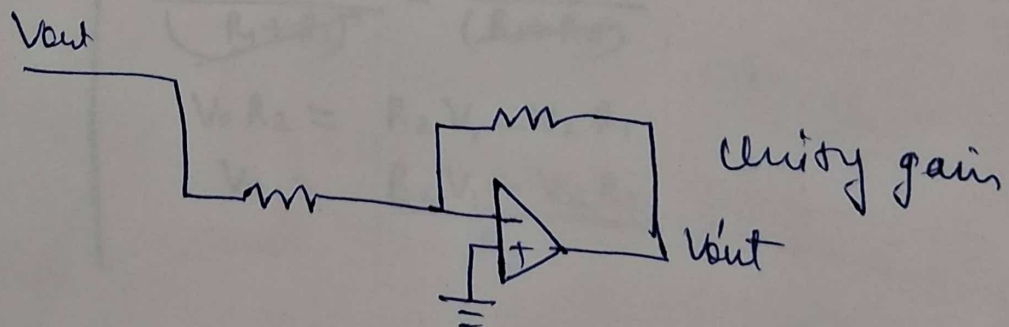
$$\Rightarrow \frac{V_o}{R_f} = -\frac{(V_1 + V_2 + V_3)}{R}$$

If $R_f = R$, then

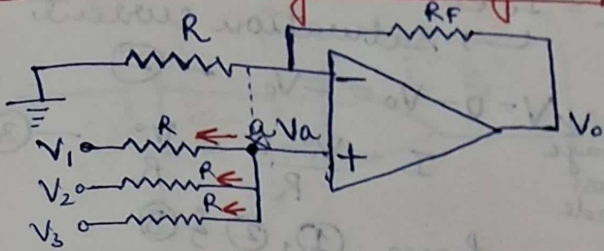
$$V_o = -(V_1 + V_2 + V_3)$$

phase shift of 180°

to make positive or non-inverting



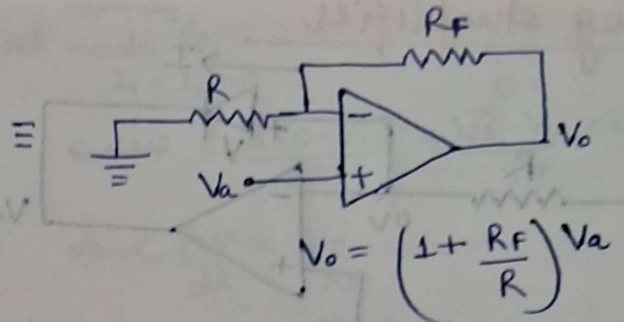
6.) Non-Inverting Summing Amplifier



KCL at a

$$\frac{V_a - V_1}{R} + \frac{V_a - V_2}{R} + \frac{V_a - V_3}{R} = 0$$

$$V_a = \frac{V_1 + V_2 + V_3}{3}$$

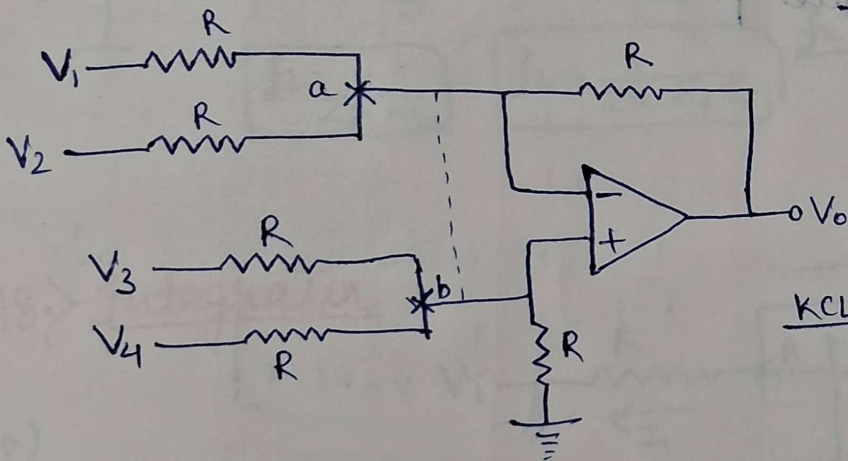


$$V_o = \left(1 + \frac{R_F}{R}\right) V_a$$

$$V_o = \left(1 + \frac{R_F}{R}\right) \left(\frac{V_1 + V_2 + V_3}{3}\right)$$

for summing $\left| \Rightarrow \frac{R_F}{R} = 2 \right.$
 $1 + \frac{R_F}{R} = 3$

7.) Adder / Subtractor



KCL at a: $\frac{V_a - V_1}{R} + \frac{V_a - V_2}{R} + \frac{V_a - V_0}{R} = 0$

$$\Rightarrow V_a = \frac{V_1 + V_2 + V_0}{3}$$

KCL at b: $\frac{V_b - V_3}{R} + \frac{V_b - V_4}{R} + \frac{V_b}{R} = 0$

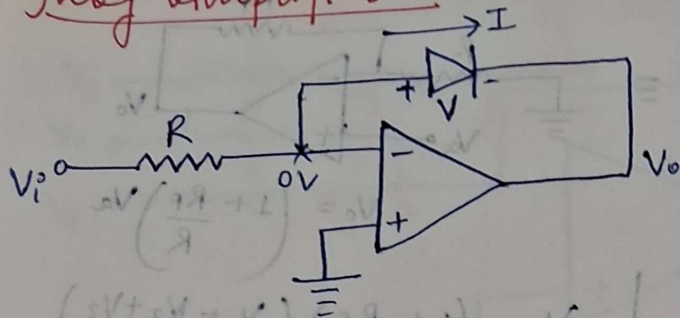
$$\Rightarrow V_b = \frac{V_3 + V_4}{3}$$

$\therefore V_a = V_b$ (Virtual Short)

$$\frac{V_3 + V_4}{3} = \frac{V_1 + V_2 + V_0}{3}$$

$$\Rightarrow V_o = (V_3 + V_4) - (V_1 + V_2)$$

8.) Log Amplifier



$$I = I_0 e^{V/nV_T} \quad \text{--- (1)}$$

↪ saturation current

$$V = 0 - V_o = -V_o \quad \text{--- (2)}$$

Voltage across diode

$$I = \frac{V_i - 0}{R} = \frac{V_i}{R} \quad \text{--- (3)}$$

from (1), (2) & (3)

$$\frac{V_i}{R} = I_0 e^{V_i/nV_T}$$

$$\frac{V_i}{R} = I_0 e^{-V_o/nV_T}$$

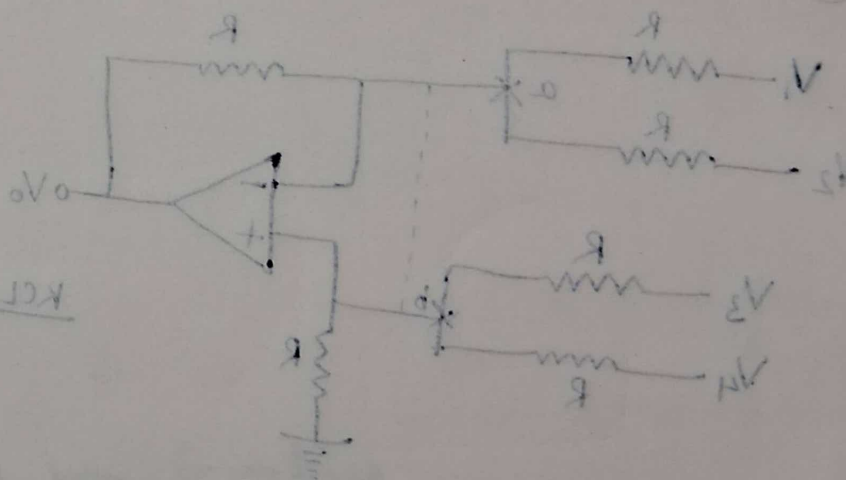
$$\ln \frac{V_i}{I_0 R} = \frac{-V_o}{nV_T} \Leftrightarrow \frac{V_i}{I_0 R} = e^{-V_o/nV_T}$$

⇓

$$V_o = -nV_T \ln \frac{V_i}{I_0 R} \Rightarrow V_o \propto \ln(KV_i)$$

(\$V_i\$ should be 'true')

* A transistor can also be used in place of diode

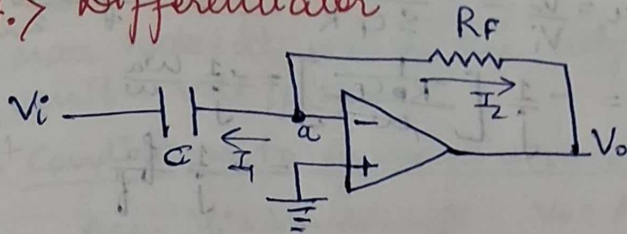


$$(V_o + V_i) - (V_o + V_i) = 0V$$

$$(V_o + V_i) - (V_o + V_i) = 0V \therefore$$

$$V_o + V_i = V_o + V_i$$

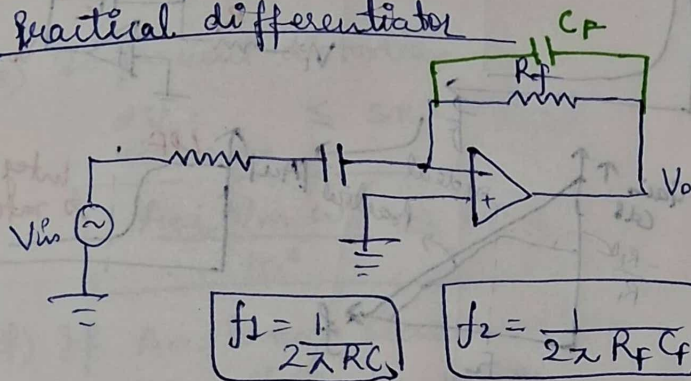
17. > Differentiator



$$\rightarrow \text{Gain} = A_{cl} = \frac{V_o}{V_i} = -\frac{R_F}{1 + sC R_F} = -\frac{s R_F C}{s \tau}$$

$$\rightarrow s = j\omega, A_{cl} = -j\omega R_F C = -R_F \cdot 2\pi f C$$

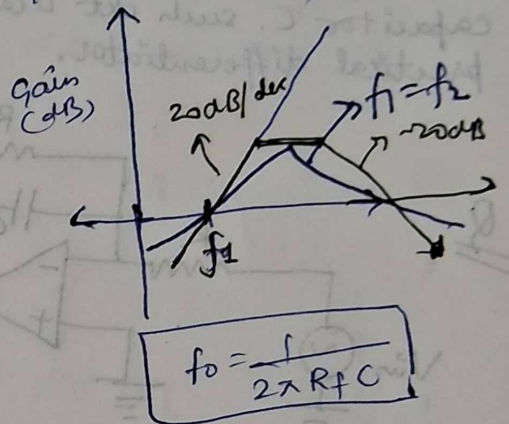
practical differentiator



$$f_1 = \frac{1}{2\pi R C}$$

$$f_2 = \frac{1}{2\pi R_F C_F}$$

$$\begin{aligned} f_1 &\neq f_2 \\ f_0 &< f_1 \text{ to } f_2 \\ f_s &= \frac{f_1}{10} \\ f_2 &> f_1 \end{aligned}$$



at node A, $I_1 + I_2 = 0$

$$C \frac{d(0 - V_i)}{dt} + \frac{0 - V_o}{R_F} = 0$$

$$\Rightarrow -C \frac{dV_i}{dt} = \frac{V_o}{R_F}$$

$$V_o = -R_F C \frac{dV_i}{dt}$$

$$V_o dt = -R_F C \int dV_i$$

$$V_o t = -R_F C V_i$$

$$\frac{V_o}{V_i} = -\frac{R_F C}{t}$$

18. > Integrator

a)

$$I_1 + I_2 = 0$$

$$\frac{0 - V_i}{R} + \frac{0 - V_o}{1/s C_F} = 0$$

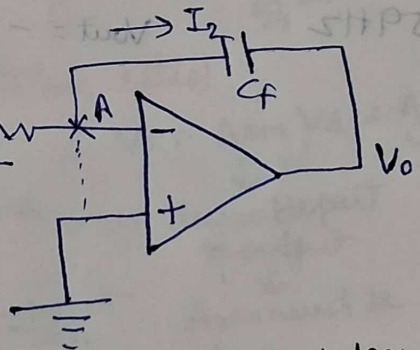
$$\frac{V_o}{V_i} = \frac{-1}{s R C_F} = -\frac{1}{R C_F} \int V_i dt$$

b) $I_1 + I_2 = 0$

$$\frac{0 - V_i}{R} + \frac{C_F d(0 - V_o)}{dt} = 0$$

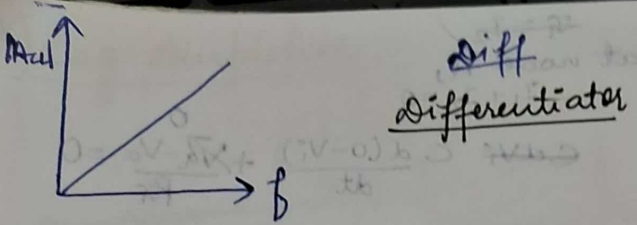
$$\frac{dV_o}{dt} = \frac{-V_i}{R C_F}$$

$$V_o = -\frac{1}{R C_F} \int V_i dt$$



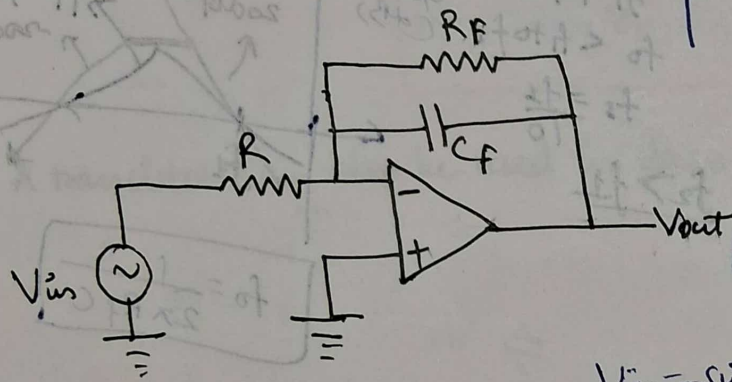
$$\left[\frac{1}{s} = \int \right]$$

V_i	Integrate	V_o



Diff
Differentiator

- If $f \rightarrow \infty$ then $|Acl| \rightarrow \infty$
- Differentiator becomes unstable at high frequencies
- produces undesired amplification to high freq noise present
- To prevent instability, a resistor can be connected in series with capacitor C. such ckt is called practical differentiator.



$$f_L = \frac{1}{2\pi R_f C_f} = 159 \text{ Hz}$$

$$10 f_L \rightarrow \frac{1}{2\pi R_f C_f}$$

$$V_{in} = \sin(2\pi 5000 t)$$

$$V_{out} = -\frac{1}{R C_f} \int V_{in}(t) dt$$

$$f_L = \frac{1}{2\pi R_f C_f}$$

$$f_0 = \frac{1}{2\pi R C_f}$$

$$f_s > 10 f_L$$

$$A_{cl} = \frac{V_o}{V_i} = -\frac{1}{s C_f R}$$

$$A_{cl} = -\frac{1}{j} \left[\frac{R C_f}{\omega} \right] = -\frac{1}{j} \frac{\omega_a}{\omega}$$

$$= -\frac{1}{j} \frac{f_a}{f}$$

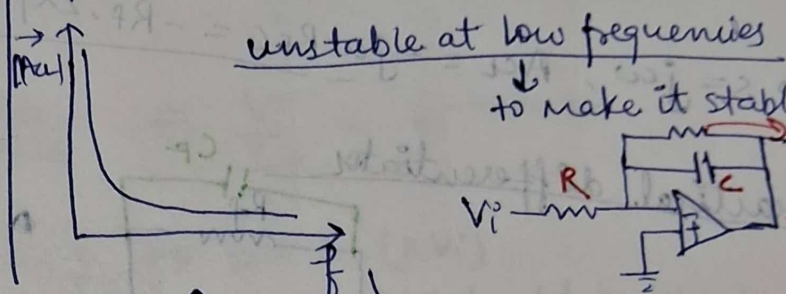
Integrator

$$2\pi f = \omega$$

$$f = \frac{\omega}{2\pi}$$

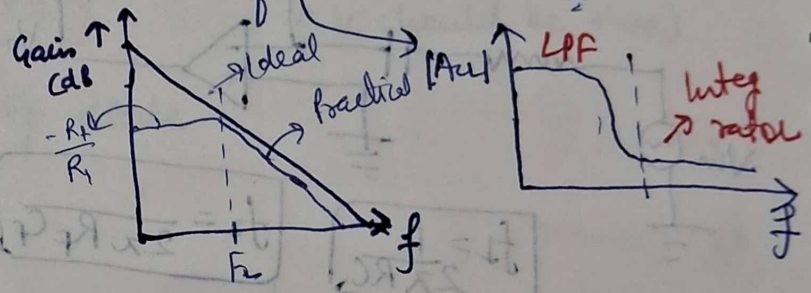
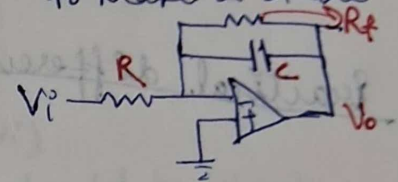
$$\rightarrow \omega_a = \frac{1}{R C_f} \text{ or } f_a = \frac{1}{2\pi R C_f}$$

$$\text{If } \omega = \omega_a, |A_{cl}| = 1$$

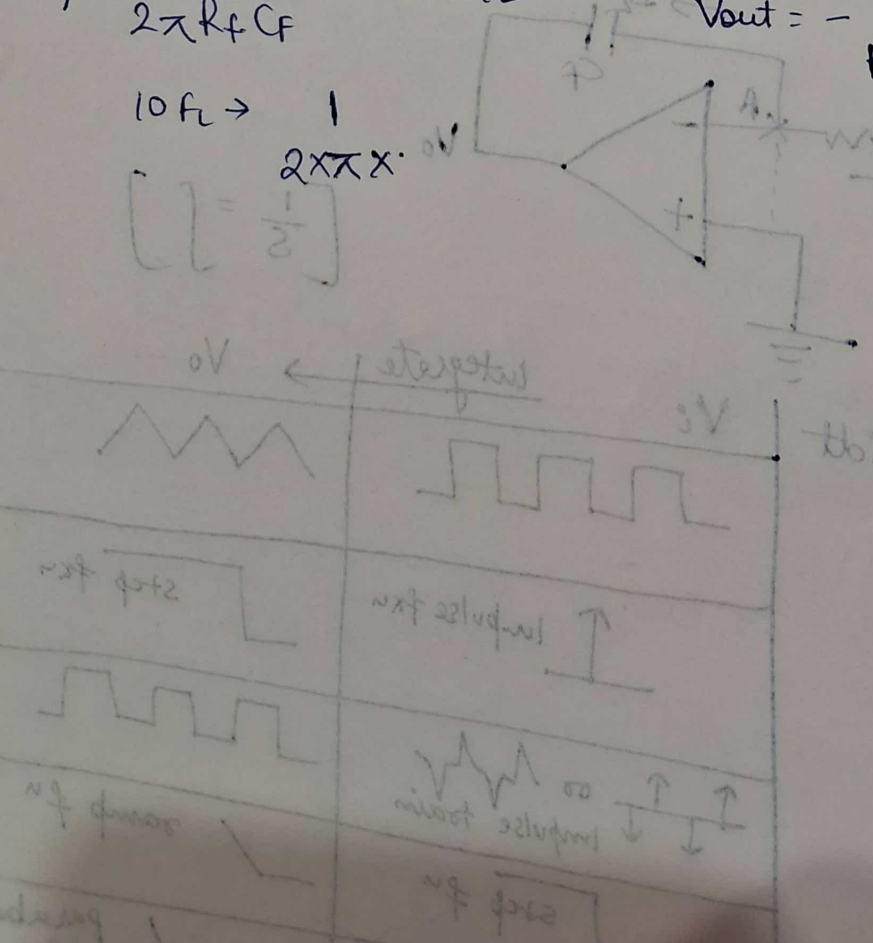


unstable at low frequencies

to make it stable



$$T = \frac{1}{f}$$



→ max rate at which the output voltage of op-amp can change.
unit: volt/ μ sec

→ Conditions to prevent distortion

a) $A_{cl} \cdot V_i = V_m \sin \omega t \quad V_o = A_{cl} V_i$

b) $V_o = A_{cl} V_m \sin \omega t$

c) $\frac{dV_o}{dt} = A_{cl} V_m \cos \omega t \cdot \omega$

d) $\left. \frac{dV_o}{dt} \right|_{\max} = \frac{A_{cl} V_m \omega}{10^6} \frac{V}{\mu \text{sec}}$

e) To prevent distortion

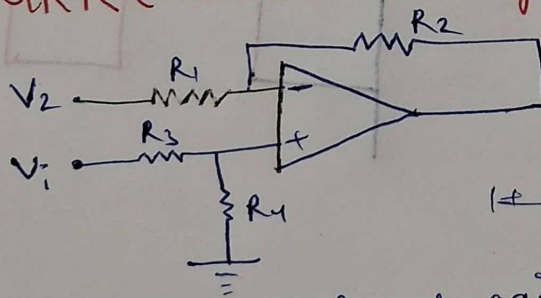
$$\left. \frac{dV_o}{dt} \right|_{\max} \leq SR \text{ (Slew Rate)}$$

$$\boxed{\frac{A_{cl} V_m \omega}{10^6} \leq SR}$$

*) If A_{cl} is not mentioned then consider it to be unity.

g) $SR \geq \left. \frac{dV_o}{dt} \right|_{\max}$

CMRR (Common mode rejection ratio)



$$V_o = A_{DM} V_d + A_{CM} V_{cm}$$

↓
desirable part
of output
↓
 A_{DM} must be
large

↓
undesirable
↓
 A_{CM} must be small

$A_{DM} \rightarrow$ Differential mode gain

$A_{CM} \rightarrow$ Common mode gain

$V_d \rightarrow$ Difference input

$V_{cm} \rightarrow$ Common mode input

A_{DM} gain when only V_1 is applied
 $A_{CM} \rightarrow$ " " " V_2 " "

$$CMRR = \frac{A_{DM}}{A_{CM}}$$

↓
must be high

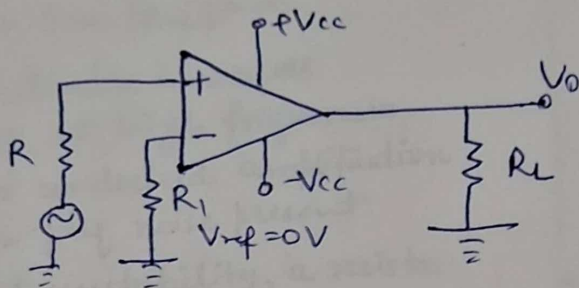
• $V_d = V_1 - V_2$

• $V_{cm} = \frac{V_1 + V_2}{2}$

• $A_{DM} = \frac{A_1 - A_2}{2}$

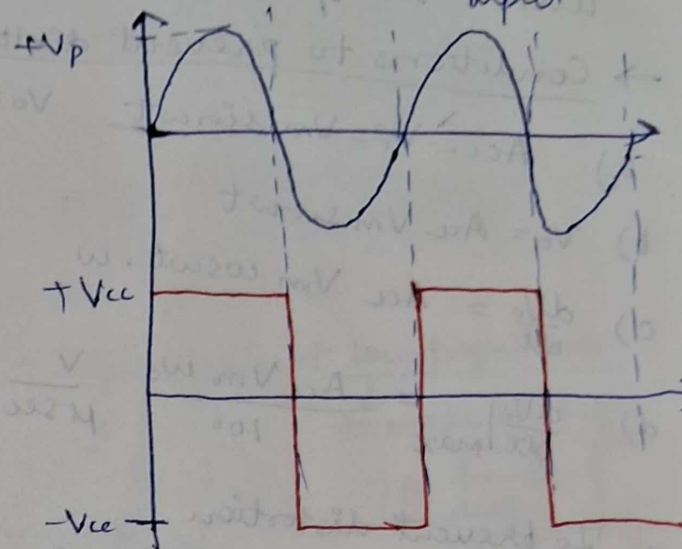
• $A_{CM} = \frac{A_1 + A_2}{2}$

a) Non-Inverting ZCD

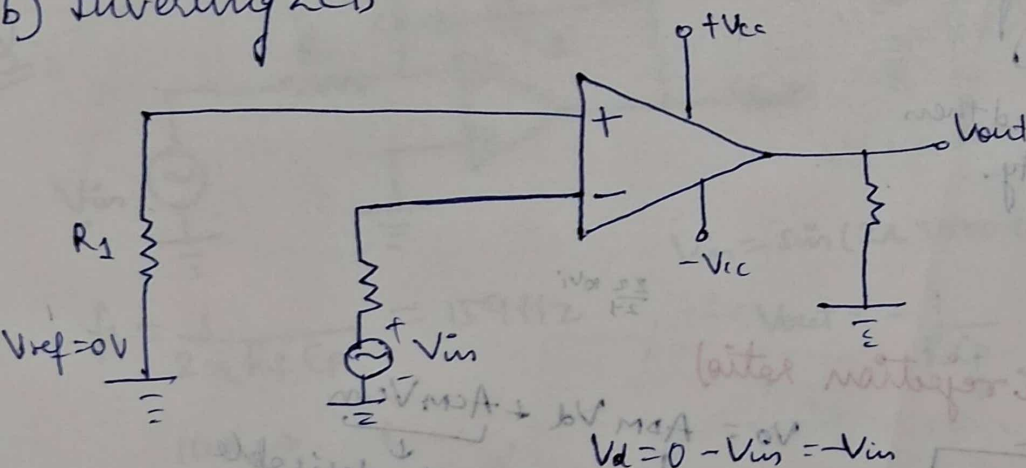


$$V_d = V_{in} - 0 = V_{in}$$

open loop $\Rightarrow$ saturation



b) Investing ZCD



$$V_d = 0 - V_{in} = -V_{in}$$

